

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/320745019>

A bi-level model formulation for the distributed wood supply planning problem

Article in *Canadian Journal of Forest Research* · October 2017

DOI: 10.1139/cjfr-2017-0240

CITATIONS

14

READS

357

4 authors:



Gregory Eric Paradis

University of British Columbia

30 PUBLICATIONS 204 CITATIONS

[SEE PROFILE](#)



Mathieu Bouchard

Université Laval

30 PUBLICATIONS 536 CITATIONS

[SEE PROFILE](#)



Luc Lebel

Université Laval

187 PUBLICATIONS 1,647 CITATIONS

[SEE PROFILE](#)



Sophie D'Amours

Université Laval

190 PUBLICATIONS 4,490 CITATIONS

[SEE PROFILE](#)

A bi-level model formulation for the distributed wood supply planning problem

Gregory Paradis, Mathieu Bouchard, Luc LeBel, and Sophie D'Amours

Abstract: The classic wood supply optimisation model maximises even-flow harvest levels and implicitly assumes infinite fibre demand. In many jurisdictions, this modelling assumption is a poor fit for actual fibre consumption, which is typically a subset of total fibre allocation. Failure of the model to anticipate this bias in industrial wood fibre consumption has been linked to increased risk of wood supply failure. In particular, we examine the distributed wood supply planning problem where the roles of forest owner and fibre consumer are played by independent agents. We use game theory to frame interactions between public forest land managers and industrial fibre consumers. We show that the distributed wood supply planning problem can be modelled more accurately using a bi-level formulation and present an extension of the classic wood supply optimisation model that explicitly anticipates industrial fibre consumption behaviour. We present a solution methodology that can solve a convex special case of the problem to global optimality and compare output and solution times of classic and bi-level model formulations using a computational experiment on a realistic dataset. Experimental results show that the bi-level formulation can mitigate risk of wood supply failure.

Key words: forest management, fibre procurement, annual allowable cut, industrial consumption, bi-level model, game theory.

Résumé : La formulation classique du modèle d'optimisation de la possibilité forestière maximise le rendement soutenu et pose l'hypothèse implicite que la demande de fibre est infinie. Dans plusieurs juridictions, cette hypothèse représente mal la réelle consommation de fibre qui est typiquement un sous-ensemble de l'attribution totale. L'incapacité du modèle classique de prévoir l'effet de ce biais entre la possibilité forestière et la consommation réelle est liée à une hausse du risque de ruptures de stock. En particulier, nous examinons le problème de planification distribuée de l'approvisionnement en bois où des agents indépendants jouent les rôles de propriétaire de la ressource forestière et de consommateur de la fibre. Nous utilisons la théorie du jeu pour encadrer les interactions entre le gestionnaire de la ressource et le consommateur industriel. Nous démontrons que le problème de planification distribuée de l'approvisionnement en bois peut être modélisé plus justement avec une formulation biniveau, et nous présentons une extension du modèle d'optimisation classique pouvant anticiper le comportement du consommateur de fibre industriel. Nous présentons une méthodologie pour résoudre un cas spécial convexe à l'optimalité globale, et nous comparons les performances expérimentales des formulations classiques et biniveaux à l'aide d'un jeu de données synthétique réaliste. Les résultats expérimentaux montrent que le modèle biniveau peut atténuer le risque de ruptures de stock.

Mots-clés : planification forestière, approvisionnement en fibre, possibilité annuelle de coupe, consommation industrielle, modèle biniveau, théorie du jeu.

1. Introduction

In Canada, provincial government authorities allocate timber licences (TLs) to industrial fibre consumers. These TLs, which grant rights to harvest fibre on public forest land, set species-wise upper bounds on periodic harvest volume; however, there is typically no policy requirement to set matching lower bounds. The TLs are typically valid for a pre-determined period (e.g., 5 years), after which point licences may be renewed, subject to re-evaluation of available wood supply. Maximal sustainable TL volume is commonly referred to as annual allowable cut (AAC).

The process by which AAC is determined is referred to as the wood supply planning problem (WSPP). We focus on a particular problem variant, which we call the distributed wood supply planning problem (DWSPP), where the roles of forest land owner and

industrial fibre consumer are played by independent agents. The DWSPP is common in Canada and other jurisdictions, where government planners are responsible for determining AAC and industrial fibre consumers are responsible for planning the actual harvesting activities.

In practice, AAC is typically determined by solving a linear programming (LP) optimisation model, which finds the maximum sustainable harvest level for a given forest (Davis et al. 2001; Gunn 2007). We will refer to this LP optimisation model formulation as the classic wood supply model.

The concept of sustainable forest management (SFM) has long been a central theme of public forest policy in Canada (Canadian Council of Forest Ministers 2008). According to the *Criteria and Indicators of Sustainable Forest Management in Canada* (CCFM C&I)

Received 20 June 2017. Accepted 14 August 2017.

G. Paradis* and L. LeBel. Département des sciences du bois et de la forêt, Faculté de foresterie, de géographie et de géomatique, Pavillon Abitibi-Price, Université Laval, Québec, QC G1K 7P4, Canada.

M. Bouchard. Consortium de recherche FORAC, Pavillon Adrien-Pouliot, Université Laval, Québec, QC G1K 7P4, Canada.

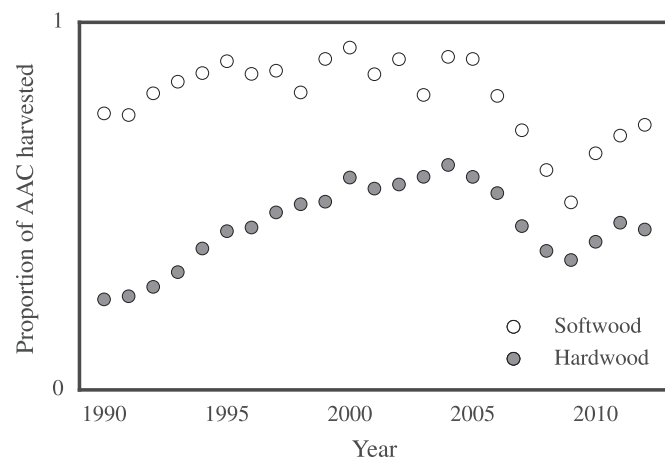
S. D'Amours. Département de génie mécanique, Faculté de sciences et de génie, Pavillon Adrien-Pouliot, Université Laval, Québec, QC G1K 7P4, Canada.

Corresponding author: Gregory Paradis (email: gregory.paradis@ubc.ca).

*Present address: Department of Forest Resources Management, University of British Columbia, 2424 Main Mall, Vancouver, BC V6T 1Z4, Canada.

Copyright remains with the author(s) or their institution(s). Permission for reuse (free in most cases) can be obtained from [RightsLink](https://www.nrcresearchpress.com/cjfr).

Fig. 1. Proportion of species-wise AAC from public forests consumed in Canada for period 1990 to 2012. Source: [Canadian Council of Forest Ministers \(2014\)](#).



([Canadian Council of Forest Ministers 2003](#)), harvesting is generally deemed to be sustainable if it is below AAC. This notion, that any harvest level below AAC is sustainable, is also implicitly used at the forest management unit (FMU) scale when government uses species-wise AAC volumes as contractual upper bounds in TLs.

The classic model simulates a finite alternating sequence of harvesting and growth and implicitly assumes that all available fibre will be consumed in every planning period (regardless of operability, quantity, quality, cost, or value creation potential). In practice, this assumption is rarely respected. [Figure 1](#) shows the species-wise proportion of Canadian AAC consumed from 1990 to 2012. On average, 80% of softwood AAC and 45% of hardwood AAC were consumed, indicating a clear industrial preference for softwood during this period. The classic model always simulates harvesting of the entire AAC in every planning period, ignoring local constraints on processing capacity of certain types of fibre, fibre procurement cost, primary breakdown cost, and market price for primary harvested wood products. Thus, the classic model implicitly assumes infinite fibre demand. This assumption is unrealistic in many cases, which explains why historical harvest levels are systematically lower than AAC. This bias is related to the infinite-demand assumption implicitly embedded in the classic wood supply optimisation model and can, to a certain extent, be attributed to a poor alignment between industrial fibre demand and wood supply planning.

Local industrial processing capacity may be insufficient to consume some parts of AAC. Other parts of AAC may be economically unattractive ([Mathey et al. 2009](#)) or be operationally inaccessible. Thus, the optimal solution of the classic model will likely never be executed, and the long-term state of the forest will be systematically different from that predicted by the wood supply model. If this systematic difference between predicted and actual outcomes is not entirely corrected by regular rolling-horizon AAC re-planning, then the current policy may not be achieving the best possible compromise between realizing short-term value-creation opportunities and maintaining long-term fibre supply stability and forest productivity.

[Paradis et al. \(2013\)](#) simulate repeated wood supply planning cycles, modelling interaction between government wood supply planners and industrial fibre consumers. They show that the classic wood supply model may fail due to the aforementioned species-skewed negative fibre consumption bias and conclude that the wood supply planning process currently in place on public land in Canada may be either overly constraining short-term value-creation opportunities, or not providing credible assurance of the long-term sustainability of the wood supply. Given the pervasive-

ness of this bias in practice, the classic wood supply optimisation model formulation does not constitute a rational basis for the implementation of sustainable forest management. We endeavor to eliminate this bias from the optimisation model used to determine AAC, thereby improving credibility of the wood supply planning process.

In many jurisdictions, sustainable forest management is implemented using a hierarchical planning framework ([Lammerts van Bueren and Blom 1997](#); [Tittler et al. 2001](#)). It has been common practice for several decades that planning tools and forestry decision support systems be designed around the hierarchical planning paradigm ([Weintraub and Cholaky 1991](#); [Weintraub et al. 2007](#); [Davis et al. 2001](#)). The notion of hierarchical planning was first formally developed in a manufacturing context, under the designation of hierarchical production planning (HPP) ([Bitran and Hax 1977](#)). In its original manufacturing context, HPP was typically implemented within a centralised decision-making environment. Effective implementation of HPP includes both coherent linkages to lower levels and effective feedback loops to higher levels to ensure coherent disaggregation of information from upper through to lower planning levels. In a centrally managed organisation, goals should be well aligned across planning levels, and lower planning levels are typically subordinate to upper planning levels. In this context, there is clear incentive to implement proper linkages and feedback loops in the HPP process.

In a distributed planning problem (DPP) ([Schneeweiss 2003](#)) environment, where independent agents with imperfectly aligned goals are responsible for different levels of the planning hierarchy, the incentive to implement proper linkages and feedback loops may be weakened or absent, as is the case for hierarchical forest management planning on public land. Despite the widespread and long-standing use of hierarchical planning as a conceptual framework for forest management planning ([Barros and Weintraub 1982](#); [Gunn and Rai 1987](#); [Weintraub and Davis 1996](#)), the linkages between long- and short-term forest management planning remain weak, and adaptive feedback loops are essentially nonexistent. These deficiencies, which are linked to failure to recognise the distributed nature of the forest management problem, may compromise the effectiveness of the planning process in achieving its stated goals. We can gain further insight into these issues by analyzing the relationships that bind these government and industry actors from a game-theoretic perspective.

[Beaudoin et al. \(2010\)](#) successfully used an agent-based modelling approach to anticipate interaction between participants in a distributed fibre procurement process at the tactical-operational planning level. They leverage this anticipation to better coordinate the planning process between horizontally linked agents (i.e., firms sourcing fibre from the same procurement area in the public forest), resulting in local and global profitability gains relative to the ad hoc base case procurement planning process. We conjecture that a sequential leader-follower anticipation approach could help coordinate the vertical integration between government and industry planners, at a strategic-tactical planning level, in an effort to close the gap between planned and executed harvest volume in the context of the DWSPP. From a game-theoretic perspective, this type of sequential leader-follower anticipation corresponds to a Stackelberg game ([von Stackelberg 2011](#)).

A Stackelberg game is a noncooperative game with two players who move sequentially. One player (the leader) moves first, and the other player (the follower) moves second. The follower can observe the leader's move before selecting his move, and the leader knows ex ante that this will be the case.

We can describe the DWSPP, from a game-theoretic perspective, in terms of a Stackelberg leader-follower game. The role of leader is played by government planners, and the role of follower is played by industrial fibre consumers. The leader has the responsibility to ensure long-term integrity of the ecosystem and the wood supply and aims to maximise AAC. The follower aims to

maximise short-term profit by transforming an optimal subset of the wood supply into forest valuable products.

From an operations research perspective, a Stackelberg game can be formulated as a bi-level planning problem (BLPP) (Colson et al. 2007). Hence, the DWSPP leader-follower problem can be formulated as a bi-level optimisation problem. We developed a bi-level model formulation for the DWSPP and a novel methodology to solve a special case of the bi-level problem to global optimality.

We hypothesized that the bi-level model formulation would improve stability of the wood supply. To verify this hypothesis, we designed a computational experiment comparing output from classic and bi-level model formulations after 30 sequential rolling-horizon re-planning cycles.

The remainder of this paper is organised as follows. Mathematical problem formulations, solution methodology, and experimental methods are presented in Section 2. Results from the computational experiment are presented in Section 3, followed by discussion in Section 4 and concluding remarks in Section 5.

2. Methods

A BLPP can be described in terms of upper- and lower-level problems (Dempe et al. 2015). We present mathematical formulations for upper- and lower-level problems of the DWSPP in Sections 2.1 and 2.2, which we combine into a bi-level formulation in Section 2.3. We describe a solution methodology to solve the bi-level model in Section 2.4, and a computational experiment comparing the performance of classic and bi-level model formulations in Section 2.5.

2.1. Upper-level problem formulation

The following formulation of the upper-level problem of the DWSPP is an adaptation of the formulation presented in Paradis et al. (2013). This is essentially a general form of the de facto standard wood supply optimization model that is used extensively in Canada and elsewhere to determine species-wise maximum even-flow AAC levels. Model input includes a description of the initial forest area (stratified by stand type and age class), growth and yield curves (linked to stand types), a set of treatments (with complex eligibility criteria), state transition matrix (keyed on stand state and treatment), and some aggregate outputs used to define the objective function and constraints.

$$\begin{aligned}
 (1) \quad & \max \sum_{i \in Z} \sum_{j \in J_i} c_{ij} x_{ij} \\
 (2) \quad & \text{s.t. } \sum_{j \in J_i} x_{ij} = 1 \quad \forall i \in Z \\
 (3) \quad & y_{p1} \leq \sum_{i \in Z} \sum_{j \in J_i} \mu_{ijp1} x_{ij} \leq (1 + \varepsilon_p) y_{p1} \quad \forall p \in O', t \in T \\
 (4) \quad & v_{ot}^- \leq \sum_{i \in Z} \sum_{j \in J_i} \mu_{ijot} x_{ij} \leq v_{ot}^+ \quad \forall o \in O, t \in T \\
 (5) \quad & 0 \leq x_{ij} \leq 1 \quad \forall i \in Z, j \in J_i
 \end{aligned}$$

where

$$\begin{aligned}
 Z &:= \text{set of spatial zones} \\
 J_i &:= \text{set of available prescriptions for zone } i \in Z \\
 O &:= \text{set of forest outputs} \\
 O' \subseteq O &:= \text{set of targeted forest outputs} \\
 T &:= \text{set of time periods in the planning horizon} \\
 \varepsilon_p &:= \text{admissible level of variation on yield of targeted output } p \in O' \\
 \mu_{ijot} &:= \text{quantity of output } o \in O \text{ produced in period } t \in T \text{ by prescription } j \in J_i \text{ in zone } i \in Z
 \end{aligned}$$

$$\begin{aligned}
 v_{ot}^- &:= \text{lower bound on yield of output } o \in O \text{ in period } t \in T \\
 v_{ot}^+ &:= \text{upper bound on yield of output } o \in O \text{ in period } t \in T \\
 c_{ij} &:= \text{objective function contribution of prescription } j \in J_i \text{ in zone } i \in Z \\
 x_{ij} &:= \text{proportion of zone } i \in Z \text{ on which prescription } j \in J_i \text{ is applied} \\
 y_{p1} &:= \sum_{i \in Z} \sum_{j \in J_i} \mu_{ijp1} x_{ij}, \text{ i.e., first-period harvest volume for targeted output } p \in O'
 \end{aligned}$$

The territory is divided into a set Z of spatial zones. For every zone $i \in Z$, we have a set of prescriptions J_i , where each prescription is a sequence of forest operations to be applied to the stands in the zone for the entire long-term planning horizon.

The objective function (eq. 1) maximises harvest volume over the planning horizon. Variables x_{ik} are linear, with domain $\{x_{ik} \in \mathbb{R} \mid 0 \leq x_{ik} \leq 1\}$ specified in eq. 5. These variables are aggregated into y_{p1} . The set $O' \subseteq O$ represents targeted forest outputs (i.e., species groups for which we determine AAC and enforce even-flow constraints). The constraint in eq. 2 requires prescriptions applied to a zone to cover the entire territory. Note that doing nothing for the entire planning horizon is considered a prescription that could generate some outputs (e.g., carbon flows, wildlife habitat, old-growth forest area, etc.). The constraint in eq. 3 enforces an even-flow policy on targeted outputs. We have expressed the even-flow constraint in terms of y_{p1} , which represents first-period harvest volume for output $p \in O'$. The constraint in eq. 4 sets upper and lower bounds on periodic yield for forest outputs. If we set $c_{ij} = \sum_{p \in O'} \sum_{t \in T} \mu_{ijpt}$, the upper-level objective function maximises targeted output harvest volume. The symbol y_{p1} corresponds to the maximum wood supply, for a given species group, that the upper level might offer the lower level in a given planning cycle — thus, y_{p1} constitutes the primary linkage interface between upper- and lower-level models.

Periodic harvest volume is stabilized using even-flow constraints, which is common in practice. The objective of the optimisation problem is to maximise species-wise AAC over the planning horizon. This corresponds to the status quo strategic planning model in many jurisdictions.

2.2. Lower-level problem formulation

The following formulation of the lower-level problem of the DWSPP is an adaptation of a multi-period problem formulation presented in Jerbi et al. (2012). The lower-level problem essentially emulates industrial fibre-consumption behaviour using a linear network flow optimization model formulation. The model formulation is very general and, as such, can accommodate a wide range of network configurations. The network is made up of a number of business units (processing nodes) that consume one or more products and produce one or more products. Input to each processing node can originate from the forest (i.e., raw logs) or from other processing nodes (e.g., chips or sawdust from a sawmill going to a pulp mill). Processes consume set proportions of input products (several inbound arcs can supply the same product) and output set mixes of output products (several outbound arcs can ship the same product). Processes also consume capacitated resources, which can limit flow through a node if a resource becomes saturated. The network features special (forest) source nodes and (market) sink nodes that essentially interface the network with upstream and downstream activities.

$$(6) \quad \max \sum_{u \in U} \left(\sum_{p \in P} \rho_{up} D_{up} - \sum_{i \in Z} \sum_{p \in O'} \pi_{iup} \beta_{iup} - \sum_{w \in W_u} c_w Y_{uw} \right) - \sum_{e \in E} \sum_{p \in P} \sigma_{ep} F_{ep}$$

$$\begin{aligned}
(7) \quad & \text{s.t. } \beta_{iup} + \sum_{w \in W_u} (\gamma_{pw} - \alpha_{pw}) Y_{uw} + \sum_{e \in \delta_u^+} F_{ep} \\
& - \sum_{e \in \delta_u^-} F_{ep} - D_{up} = 0 \quad \forall i \in Z, u \in U, p \in P \\
(8) \quad & \sum_{w \in W_u} \lambda_{kuw} Y_{uw} \leq q_{ku} \quad \forall u \in U, k \in K \\
(9) \quad & D_{up} \leq d_{up} \quad \forall u \in U, p \in P \\
(10) \quad & \sum_{p \in P} F_{ep} \leq f_e^+ \quad \forall e \in E \\
(11) \quad & f_{ep}^- \leq F_{ep} \leq f_{ep}^+ \quad \forall e \in E, p \in P \\
(12) \quad & \sum_{i \in Z} \sum_{u \in U} \beta_{iup} \leq y_{p1} \quad \forall p \in O' \\
(13) \quad & \beta_{iup} = 0 \quad \forall i \in Z, u \in U, p \in P \setminus O' \\
(14) \quad & D_{up} \geq 0 \quad \forall u \in U, p \in P \\
(15) \quad & \beta_{iup} \geq 0 \quad \forall i \in Z, u \in U, p \in P \\
(16) \quad & Y_{uw} \geq 0 \quad \forall u \in U, w \in W \\
(17) \quad & F_{ep} \geq 0 \quad \forall e \in E, p \in P
\end{aligned}$$

where

- U := set of business units
- K := set of resource capacity types (machine capacities, stock limits)
- W := set of processes (machines, inventories)
- $W_u \subseteq W$:= set of processes available at business unit $u \in U$
- O := set of outputs (from upper-level model)
- P := set of products
- $O' \subseteq P$:= set of sustainable forest outputs, from the upper-level model
- E := set of links between business units
- $\delta_u^+ \subseteq E$:= set of inbound links for business unit $u \in U$
- $\delta_u^- \subseteq E$:= set of outbound links for business unit $u \in U$
- q_{ku} := capacity of type $k \in K$ at business unit $u \in U$
- f_{ep}^- := lower bound on flow of product $p \in P$ through link $e \in E$
- f_{ep}^+ := upper bound on flow of product $p \in P$ through link $e \in E$
- f_e^+ := upper bound on flow of all products through link $e \in E$
- π_{iup} := unit cost of procuring product $p \in O'$ from forest zone $i \in Z$ to business unit $u \in U$
- c_w := unit cost of process $w \in W$
- σ_{ep} := unit cost of transporting product $p \in P$ on link $e \in E$
- α_{pw} := quantity of product $p \in P$ required for one unit of process $w \in W$
- γ_{pw} := quantity of product $p \in P$ produced for one unit of process $w \in W$
- λ_{kuw} := capacity of type $k \in K$ utilised by one unit of process $w \in W$ at business unit $u \in U$
- d_{up} := demand for product $p \in P$ at business unit $u \in U$
- ρ_{up} := price of product $p \in P$ at business unit $u \in U$
- y_{p1} := maximum external supply of sustainable forest output $p \in O'$ (from upper-level model)

and

- D_{up} := quantity of product $p \in P$ sold by business unit $u \in U$
- β_{iup} := external supply of product $p \in P$ from forest zone $i \in Z$ to business unit $u \in U$
- Y_{uw} := quantity of process $w \in W$ performed at business unit $u \in U$
- F_{ep} := flow of product $p \in P$ on link $e \in E$

The variables in the lower-level problem are given by D_{up} , β_{iup} , Y_{uw} , and F_{ep} .

Note that storage of unsold products and end-product consumption by external clients are modelled as special cases of processes. Fibre procurement from the forest is also modelled as a process, although we present it here using a separate symbol β_{iup} , as this facilitates description of the linkage with the upper-level model, via symbol y_{p1} . For simplicity, we have chosen to model industrial fibre consumption in the lower level for the first planning period only, although we could extend anticipation of industrial fibre consumption behaviour to an arbitrary number of periods without loss of generality.

The objective function in eq. 6 maximises total network profit. The constraint in eq. 7 enforces flow conservation in the network. The constraint in eq. 8 models process capacity constraints. The constraint in eq. 9 ensures that sale of products to end-customers does not exceed demand. The constraints in eqs. 10 and 11 enforce total and product-wise link flow constraints. The constraint in eq. 12 ensures flow of fibre from the forest does not exceed species-wise upper bounds from the upper-level model. The constraint in eq. 13 ensures null external supply for any product $p \in P$ that is not in set O' . The constraints in eqs. 14–17 are non-negativity constraints on decision variables.

2.3. Bi-level problem formulation

Having formulated the optimisation problems of the upper and lower levels, we can combine both levels to form a BLPP. Recall that the motivation for modelling the DWSPP as a BLPP is to anticipate industrial fibre consumption in the wood supply model, thereby eliminating the species-skewed gap between AAC and the volume of fibre actually consumed by industry. The bi-level formulation for the DWSPP is given by

$$\begin{aligned}
(18) \quad & \max \sum_{i \in Z} \sum_{j \in J_i} c_{ij} x_{ij} \\
(19) \quad & \text{s.t. } \sum_{j \in J_i} x_{ij} = 1 \quad \forall i \in Z \\
(20) \quad & y_{p1} \leq \sum_{i \in Z} \sum_{j \in J_i} \mu_{ijpt} x_{ij} \leq (1 + \varepsilon_p) y_{p1} \quad \forall p \in O', t \in T \\
(21) \quad & v_{ot}^- \leq \sum_{i \in Z} \sum_{j \in J_i} \mu_{ijot} x_{ij} \leq v_{ot}^+ \quad \forall o \in O, t \in T \\
(22) \quad & y_{p1} \leq \sum_{i \in Z} \sum_{u \in U} \beta_{iup} \quad \forall p \in O' \\
(23) \quad & 0 \leq x_{ij} \leq 1 \quad \forall i \in Z, j \in J_i \\
(24) \quad & \beta_{\bar{p}} \in \arg\max_{\beta_{\bar{p}}} \sum_{u \in U} \left(\sum_{p \in P} \rho_{up} D_{up} - \sum_{i \in Z} \pi_{iup} \beta_{iup} \right. \\
& \quad \left. - \sum_{w \in W_u} c_w Y_{uw} \right) - \sum_{e \in E} \sum_{p \in P} \sigma_{ep} F_{ep} \quad \forall \bar{p} \in O' \\
(25) \quad & \text{s.t. } \beta_{iup} + \sum_{w \in W_u} (\gamma_{pw} - \alpha_{pw}) Y_{uw} + \sum_{e \in \delta_u^+} F_{ep} \\
& \quad - \sum_{e \in \delta_u^-} F_{ep} - D_{up} = 0 \quad \forall i \in Z, u \in U, p \in P \\
(26) \quad & \sum_{w \in W_u} \lambda_{kuw} Y_{uw} \leq q_{ku} \quad \forall u \in U, k \in K
\end{aligned}$$

$$(27) \quad D_{up} \leq d_{up} \quad \forall u \in U, p \in P$$

$$(28) \quad \sum_{p \in P} F_{ep} \leq f_e^+ \quad \forall e \in E$$

$$(29) \quad f_{ep}^- \leq F_{ep} \leq f_{ep}^+ \quad \forall e \in E, p \in P$$

$$(30) \quad \sum_{i \in Z} \sum_{u \in U} \beta_{iup} \leq y_{p1}$$

$$(31) \quad \sum_{j \in J_i} \mu_{ijp1} \hat{x}_{ij} = \sum_{u \in U} \beta_{iup} \quad \forall i \in Z$$

$$(32) \quad \sum_{u \in U} \left(\sum_{p \in P} \rho_{up} D_{up} - \sum_{i \in Z} \pi_{iup} \beta_{iup} - \sum_{w \in W_u} c_w Y_{uw} \right) - \sum_{e \in E} \sum_{p \in P} \sigma_{ep} F_{ep} \geq 0$$

$$(33) \quad \sum_{j \in J_i} \hat{x}_{ij} = 1 \quad \forall i \in Z$$

$$(34) \quad v_{o1}^- \leq \sum_{i \in Z} \sum_{j \in J_i} \mu_{ij o1} \hat{x}_{ij} \leq v_{o1}^+ \quad \forall o \in O$$

$$(35) \quad D_{up} \geq 0 \quad \forall u \in U, p \in P$$

$$(36) \quad \beta_{iup} \geq 0 \quad \forall i \in Z, u \in U, p \in O'$$

$$(37) \quad Y_{uw} \geq 0 \quad \forall u \in U, w \in W$$

$$(38) \quad F_{ep} \geq 0 \quad \forall e \in E, p \in P$$

$$(39) \quad 0 \leq \hat{x}_{ij} \leq 1 \quad \forall i \in Z, j \in J_i$$

where $\beta_{\bar{p}}$ represents the vector of fibre flows for targeted output $\bar{p} \in O'$ from all forest zones $i \in Z$ to all units $u \in U$, and $\hat{x}_{ij}$ represents the lower-level (i.e., as-implemented) harvest solution.

The upper-level objective function in eq. 18 maximises harvest volume over the planning horizon. The upper-level forest treatment variables x_{ik} are linear, with domain $\{x_{ik} \in \mathbb{R} \mid 0 \leq x_{ik} \leq 1\}$, as specified in in eq. 23. These variables are aggregated into y_{p1} . The set $O' \subseteq O$ represents targeted outputs (i.e., species groups for which we determine AAC and enforce even-flow constraints). The upper-level constraint in eq. 19 requires prescriptions applied to a zone by the upper level to cover the entire territory. The upper-level constraint in eq. 20 enforces an even-flow policy on targeted outputs. We have expressed the even-flow constraint in terms of y_{p1} . The upper-level constraint in eq. 21 bounds minimal and maximal periodic yield on the general outputs. If we set $c_{ij} = \sum_{p \in O'} \sum_{t \in T} \mu_{ijpt}$, the upper-level objective function maximises AAC. The upper-level constraint in eq. 22 ensures that first-period harvest volume can and will be entirely consumed by the profit-maximizing lower level. This constraint was not present in the upper-level model formulation presented in Section 2.1.

The lower-level objective function in eq. 24 maximises total network profit, for subproblems considering flows from each targeted output $\bar{p} \in O'$ individually. Lower-level constraint in eq. 25 enforces flow conservation in the network. Lower-level constraint in eq. 26 models process capacity constraints. Lower-level constraint in eq. 27 ensures that sale of products to end-customers does not exceed demand. Lower-level constraints in eqs. 28 and 29 enforce total and product-wise link flow bounds. Lower-level constraint in eq. 30 ensures flow of fibre from the forest does not exceed species-wise upper bounds from the upper-level model.

Lower-level constraint in eq. 31 ensures that all fibre harvested in the lower-level harvest vector $\hat{x}$ is consumed by the network. Lower-level constraint in eq. 32 ensures that profit is non-negative for the subproblem. Lower-level harvest variables $\hat{x}_{ik}$ are linear, with domain $\{\hat{x}_{ik} \in \mathbb{R} \mid 0 \leq \hat{x}_{ik} \leq 1\}$ specified in in eq. 39. Lower-level constraints in eq. 35 through eq. 38 are non-negativity constraints on lower-level flow variables. Lower-level constraint in eq. 33 requires prescriptions applied to a zone by the lower level to cover the entire territory. Lower-level constraint in eq. 34 bounds minimal and maximal first-period yield on the general outputs.

Note that symbols y_{p1} and β_{iup} appear in both upper- and lower-level parts of the bi-level model. y_{p1} is controlled by the upper level, whereas β_{iup} is controlled by the lower level. The interpretation of these symbols (i.e., as variables or parameters) depends on where they appear in the model. $\sum_{i \in Z} \sum_{u \in U} \beta_{iup}$ represents an upper bound on y_{p1} in the upper-level problem, whereas y_{p1} represents an upper bound on $\sum_{i \in Z} \sum_{u \in U} \beta_{iup}$ in the lower-level model.

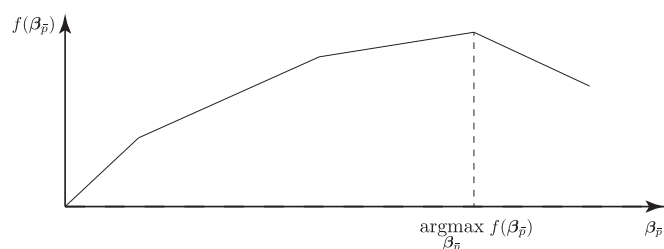
Strictly speaking, this is not a well-posed optimisation problem, as there may be multiple feasible β flow vectors and $\hat{x}$ harvest vectors that induce the same optimal objective function value in the lower-level problem, for a given $\bar{p} \in O'$. If we assume that the leader can select a single follower optimal $\hat{x}$ harvest vector that maximises his own objective function (i.e., the optimistic bi-level planning problem (OBLPP); see Dempe et al. 2015 and Paradis 2016), then the problem is well-formed and can potentially be solved to global optimality. The $\hat{x}$ harvest vector corresponds to the first-period lower-level harvest plan, which may (in the most general interpretation of the bi-level DWSP) differ from the first-period harvest plan selected by the upper level. If we set $\hat{x} = x$, this corresponds to the optimistic assumption that induces the OBLPP. Scenario 2, presented later in this paper, simulates the OBLPP (i.e., simulates the long-term outcome of the system if the optimistic assumption holds true).

The bi-level feasible region is the subset of the upper-level feasible region for which the lower-level model consumes the entire wood supply, while ensuring non-negative profit for each lower-level subproblem. In other words, a bi-level-feasible wood supply solution must be upper level feasible and be entirely consumed by the lower-level model, while ensuring non-negative profit for all species-wise lower-level subproblems.

The classic definition of AAC fails to account for the species-skewed negative fibre consumption bias (i.e., fails to account for realistic industrial fibre consumption behaviour). We propose an extended definition of AAC, which is the maximum sustainable harvest level that will be entirely consumed by a profit-maximizing network of industrial consumers. Given this extended definition of AAC, even-flow wood supply solutions that will not be entirely consumed by the follower are no longer feasible. If deterministic assumptions hold true — the assumption of determinism is implicit to linear programming, which we use to model both upper and lower levels of the problem — this extended definition will completely eliminate the problematic species-skewed negative fibre consumption bias. Although defining bi-level AAC this way is relatively straightforward, the resulting bi-level solution space is nonconvex, which makes it difficult to solve the bi-level wood supply problem to optimality.

Early model development efforts focused on what we will hereafter refer to as the general bi-level case, which we hoped to solve to global optimality using an iterative solution methodology, based on the pioneering work of Fortuny-Amat and McCarl (1981) and Bard and Falk (1982). After developing the mathematical formulation for the general bi-level case, we recognised the potential for nonconvexity of the bi-level solution space. This limited the worst-case performance of our initial solution methodology approach to local optimal solutions. Given the strategic importance of wood supply planning, we considered local optimal solutions to be of limited interest in the absence of reliable bounds on the

Fig. 2. Illustration of concave profit function of a hypothetical separable output $\bar{p}$.



optimality gap. Thus, we decided to focus subsequent efforts on identifying special cases of the bi-level problem that we could solve to global optimality.

Nonconvex cases can occur when the lower-level model features one or more convergent processes with saturated capacity constraints. A convergent process accepts input derived from two or more sustainable forest outputs $\bar{p} \in O'$. For example, this could occur if both hardwood and softwood chips in a network must flow through a single pulp-mill (i.e., convergent process), with a single digester that runs nonstop (i.e., shared resource with saturated capacity constraints).

By restricting the problem domain to nonproblematic (i.e., convex) instances, we can define a special case of the bi-level problem that can be solved more easily. The special case, which is defined by excluding problematic instances from the general problem domain, occurs in two types of instances. Special case 1 corresponds to strictly divergent networks (i.e., no convergence of product flows at any process) and can be easily identified by analyzing network topology of the lower-level model. Special case 2 corresponds to partially convergent networks with sufficient capacity at convergent processes so these will not be saturated at the bi-level optimal solution. Special case 2 may only be observable through empirical analysis of the lower level model and may not be detectable a priori, as checking for potential capacity constraint saturation requires us to first solve the lower level model.

Both types of species cases are separable in O' (i.e., the species groups used to segment AAC in the upper level model). Thus, we can develop a solution methodology for the special case that decomposes the lower-level problem into convex species-wise subproblems, which can be solved to determine optimal species-wise upper bounds on harvest levels in the upper-level problem. These species-wise upper bounds represent valid cuts for the upper-level solution space. These cuts form the basis for our solution methodology to solve the special cases. We cannot generate similar valid cuts for the general case, using this methodology, due to potential non-convexity of the general bi-level DWSPP solution space.

The species-wise upper-bound volume for output $\bar{p} \in O'$ (i.e., solution to lower level subproblem for output $\bar{p}$) is equal to $\arg\max_{\beta_p} f(\beta_p)$, where $f(\beta_p)$ is a concave piecewise linear function describing subproblem profit as a function of the volume of product $\bar{p}$ consumed by the lower-level model. The piecewise profit function is concave for each separable output $\bar{p} \in O'$, for any non-negative consumption volume $\sum_{i \in Z} \sum_{u \in U} \beta_{iup}$ (Fig. 2). At the lower extremity of the piecewise linear function domain, the follower utilizes his limited resources to produce his most profitable product mix until one of his resource constraints is saturated. The follower then switches production to the next most profitable product mix. The slope of each line segment in the piecewise function represents unit profit from a given product mix. By definition, the slopes of these line segments will be monotonically decreasing, hence the concave shape of the piecewise linear profit function.

2.4. Solution methodology

We developed a solution methodology to solve the bi-level problem presented in Section 2.3. By limiting the problem domain to

the special cases, we eliminate the possibility of interaction between outputs $\bar{p} \in O'$ (i.e., the bi-level problem becomes separable in O'). This property allows us to compute output-wise upper bounds on lower-level fibre consumption behaviour, which can be used to optimally constrain the upper-level problem to produce the global optimal bi-level solution to the DWSPP. This is the basis of our solution methodology. Algorithm 1 describes this solution methodology using pseudo-code.

Output: Global optimal wood supply solution x^*

foreach $\bar{p} \in O'$ **do**

Solve lower-level subproblem to determine upper bound $\sum_{i \in Z} \sum_{u \in U} \beta_{iup}$ on consumption of $\bar{p}$.
Set upper bound constraint $y_{p1} \leq \sum_{i \in Z} \sum_{u \in U} \beta_{iup}$ in the upper-level problem.

end

Solve constrained upper-level model.

Algorithm 1 solves the bi-level model presented in Section 2.3. The lower-level subproblems are equivalent to eqs. 2–39. The optimal solution to each lower-level subproblem yields the maximum volume $\sum_{i \in Z} \sum_{u \in U} \beta_{iup}$ of output $\bar{p}$ that we can expect the follower will consume. Conceptually, this corresponds to $\arg\max_{\beta_p} f(\beta_p)$ (see Fig. 2).

For the special case, each upper bound volume $\arg\max_{\beta_p} f(\beta_p)$ constitutes a valid cut for the bi-level solution space. Applying similar cuts to the general case would possibly exclude the global optimal solution, due to nonconvexity of the solution space.

Lower-level datasets corresponding to one or the other of the special cases can be solved to optimality using Algorithm 1. Convergent lower-level datasets that have saturated shared capacity constraints can always be transformed to special case 2 by relaxing the saturated shared capacity constraints. The test dataset used in our computational experiment, which we describe in the next section, is an example of special case 2.

2.5. Computational experiment methodology

This section describes the computational experiment we conducted to compare the performance of classic and bi-level wood supply models.

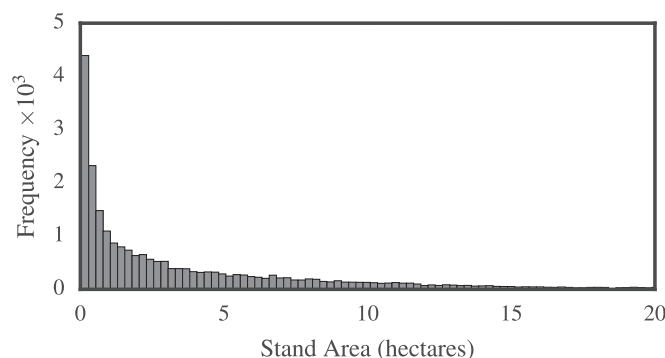
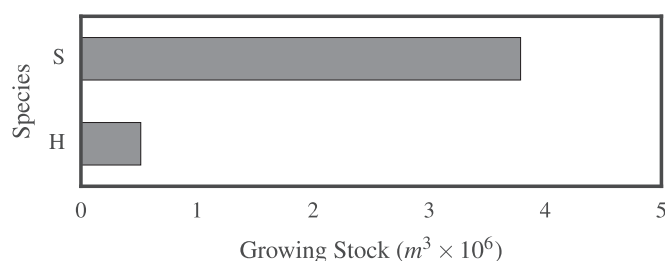
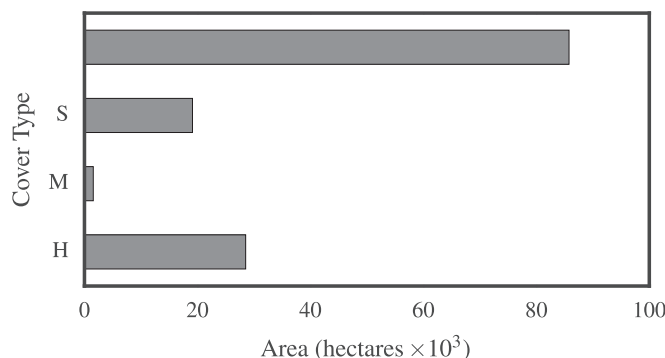
2.5.1. Test dataset

We tested our bi-level solution methodology on a realistic synthetic dataset from Quebec, Canada. The upper-level dataset represents FMU 031-53, located in the boreal forest region. The lower-level dataset represents a hypothetical network of five mills.

Our upper-level test dataset is adapted from a government wood supply model for FMU 031-53 in Quebec, Canada. We use the same dataset as that used in Paradis et al. (2013). Although the upper- and lower-level problem formulations described earlier allow the forest dataset to be subdivided into any number of zones (for example, to account for differences in transportation cost, stumpage fees, or eligibility for harvest treatments), our test dataset has only a single zone. This simplifies interpretation of our simulation results for the purposes of comparing the output of our bi-level formulation with the output of the classic model. Initial forest inventory, silviculture treatment eligibility and operability, yield curves, and state transition matrix were all compiled by government wood supply analysts for the 2013–2018 AAC planning period.

The original dataset has a total area of 116 538 ha, 104 413 ha of which are included in the wood supply analysis (10.4% area net-down). The productive area is made up of 24 373 stands. Figure 3 shows the initial stand size distribution for our test dataset.

The test area is in the boreal forest type. The majority (88%) of initial growing stock is softwood, mostly black spruce (*Picea mariana* (Mill.) Britton, Sterns & Poggenb.) and balsam fir (*Abies balsamea* (L.) Mill.), with a presence (12%) of hardwood species, mostly white birch (*Betula papyrifera* Marshall) and trembling as-

Fig. 3. Initial stand size distribution for forest management unit 031-53.**Fig. 4.** Initial growing stock, by species group, for forest management unit 031-53. The two species groups are softwood (S) and hardwood (H).**Fig. 5.** Initial cover type distribution for forest management unit 031-53. The three cover types are softwood (S), mixedwood (M), and hardwood (H).

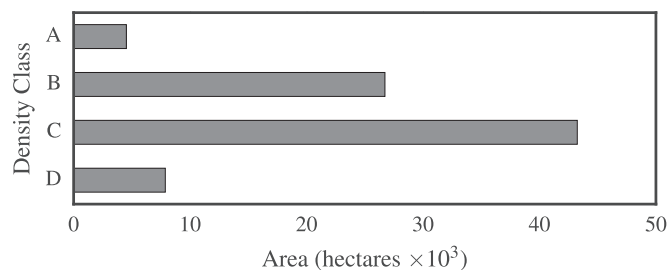
pen (*Populus tremuloides* Michx.). Figure 4 shows the initial growing stock for FMU 031-53.

Some pure spruce stands occur naturally, and plantations are generally pure spruce. A significant proportion of the forest cover is made up of mixed-wood stands containing different proportions of hardwood mixed in with the softwood. Figures 5 and 6 show the initial cover type and density class distributions of this management unit. Figure 7 shows the initial age class distribution for our test dataset.

The most recently published official AAC (determined by government planners) is 101 000 m³ for softwood and 10 000 m³ for hardwood.

The original government wood supply model features an array of silviculture treatment options. Our simplified test dataset implements one commercial treatment (clearcut) and two noncommercial treatments (planting, pre-commercial thinning). For our test model, no species-wise selective cutting is allowed (i.e., hardwood must be harvested if present) in mixed-wood stands.

For the lower-level planning model, we compiled an illustrative case study using realistic parameters (production capacities, unit cost of raw logs, transformation process input or output maps, unit prices of finished products). The case study dataset does not

Fig. 6. Initial density class distribution for forest management unit 031-53. The four density classes, expressed as a proportion of complete crown closure, correspond to A: (0.25, 0.40], B: (0.40, 0.60], C: (0.60, 0.80], and D: (0.80, 1.00].**Fig. 7.** Initial age class distribution for forest management unit 031-53.

represent an actual network of industrial fibre consumers; however, these parameters were synthesised from realistic data that were compiled for previous research projects realised by the FORAC Research Consortium. Parameter values used were originally collected from industry partners in the course of previous research projects.

The lower-level data model allows for an arbitrary number of external fibre suppliers, with each supplier potentially having distinct input parameters. Our test dataset features a single fibre procurement zone, thus all input units of fibre in the lower-level model have the same unit procurement cost. In reality, procurement cost may vary due to a number of factors (e.g., distance from mill, accessibility, terrain, choice of harvesting system and silviculture prescription, species mix, etc.). However, modelling variable procurement cost was not necessary given the objectives of our computational experiments and would only have served to obfuscate the results. We model the forest (i.e., external supply of fibre for the network) as a business unit, which encapsulates processes that convert raw fibre to an assortment of logs. The forest business units produce three types of softwood logs (small, medium, and large) and one type of hardwood log.

The three types of softwood logs can flow to any of the three softwood sawmills in the network. The softwood sawmills produce four types of softwood lumber (2x3, 2x4, 2x6, and 2x8) and softwood chips. Softwood lumber can be sold to a single softwood lumber external customer.

The hardwood logs are processed by the single hardwood sawmill, which produces a single type of hardwood lumber and hardwood chips. Hardwood lumber can be sold to a single hardwood lumber external customer.

Both hardwood and softwood chips flow from the sawmills to the paper mill, where they can be converted to paper. The paper can be sold to one of two paper external customers. There is no external customer for chips.

Output from the upper-level model is aggregated into two outputs: softwood and hardwood. The lower-level model has limited

Table 1. Summary of scenario parametres.

Scenario	Upper-level model	Lower-level model	Description
1	Classic	Basic	Control scenario (status quo)
2	Bi-level	Slave	Best-case bi-level
3	Bi-level	Basic	Base bi-level
4	Bi-level (80% attribution [†])	Basic	Reduced attribution bi-level
5	Bi-level (60% attribution [†])	Basic	Reduced attribution bi-level (stable)

[†]Proportion of bi-level softwood AAC volume determined in scenario 3.

capacity for transforming hardwood (approximately one-third of maximum biophysical AAC).

Our test dataset corresponds to an instance of special case 2. Although chip flows from both hardwood and softwood sawmills converge at the pulp mill, we have determined empirically that the shared resource capacity at the pulp mill is sufficient to avoid saturation problems.

2.5.2. Iterative simulation framework

We use the two-stage rolling-horizon simulation framework presented in Paradis et al. (2013) to compare the performance of classic and bi-level wood supply model formulations. At each simulated planning cycle, the leader and the follower make their moves sequentially, in a two-stage game. For our computational experiments, we chose to simulate 30 (5 year) re-planning cycles, as this corresponds to the length of our wood supply model planning horizon. The framework simulates forest growth between each 5-year rolling-horizon re-planning cycle. The leader has the advantage of the first move, which means he can set AAC to any level of his choosing. Algorithm 2 describes the iterative simulation algorithm using pseudo-code.

```
foreach  $t \in T$  do
    First stage: leader determines AAC (species-wise bounds on follower consumption).
    Second stage: follower consumes a subset of AAC.
    Roll planning horizon forward one period.
end
```

The simulation algorithm can be summarized as follows:

1. First stage: the leader determines his wood supply offer. We simulate the wood supply planning process by solving a wood supply optimisation model. Which model we solve at this stage — either the classic (single-level) model or the extended (bi-level) model — depends on the scenario. The wood supply offer is communicated to the follower in terms of species-wise upper bounds on volume that can be harvested in the second stage.
2. Second stage: the follower consumes a subset of the wood supply. We simulate fibre consumption by solving a network flow model, to determine the profit-maximizing subset of wood supply that the follower would willingly consume. Species-wise upper bounds from the first stage (i.e., AAC) are applied at this stage.
3. Simulate rolling horizon forward one period (simulate evolution of forest state using growth and yield curves from long-term wood supply model).

2.5.3. Experimental methodology

We present five scenarios. Within each scenario, simulation parametres for the industrial fibre consumption network are held constant for all 30 planning cycles. Table 1 summarises simulation parametres for each scenario.

Scenario 1 simulates status quo behaviour for both leader and follower and acts as a control scenario. At each planning cycle, the leader maximises even-flow AAC (30-period horizon) using the classic wood supply model, and then, the follower maximises

first-period profits (one-period horizon) by consuming an optimal subset of the wood supply offered by the leader. The leader does not consider the follower’s fibre consumption capacity when determining AAC.

Scenario 2 presents a perfect implementation of the bi-level scenario — rather than being allowed to re-plan harvesting on a one-period horizon (as is the case for all other scenarios), we force the follower to exactly implement the first period of the leader’s wood supply solution. This scenario shows the best-case performance of the bi-level model solution and corresponds to the OBLPP.

Scenario 3 is the basic bi-level scenario. The leader uses the bi-level model to determine AAC, and the follower is allowed to re-plan harvesting on a one-period horizon, choosing any profit-maximizing subset of available wood supply. Due to the optimal formulation of the bi-level model and perfect anticipation of volume consumption, the follower always chooses to harvest the entire wood supply volume. However, during the second stage of each simulation iteration, the follower may select to harvest this volume from a different combination of forest types than that which was prescribed in the first period of the leader’s optimal solution. Conceptually, the contrast between scenarios 2 and 3 is an estimate of sensitivity of long-term wood supply to deviations from the leader’s optimal bi-level wood supply plan (i.e., sensitivity to the assumption that this problem can be formulated as an OBLPP).

Scenarios 4 and 5 are variants of scenario 3, simulating reduced upper bounds on softwood supply volume. This indirectly creates a buffer stock to protect against the effects of the follower re-planning harvest decision vector $\mathbf{x}$ in the second stage of each simulation iteration (i.e., compensation for the leader’s incomplete control of the DWSPP).

3. Results

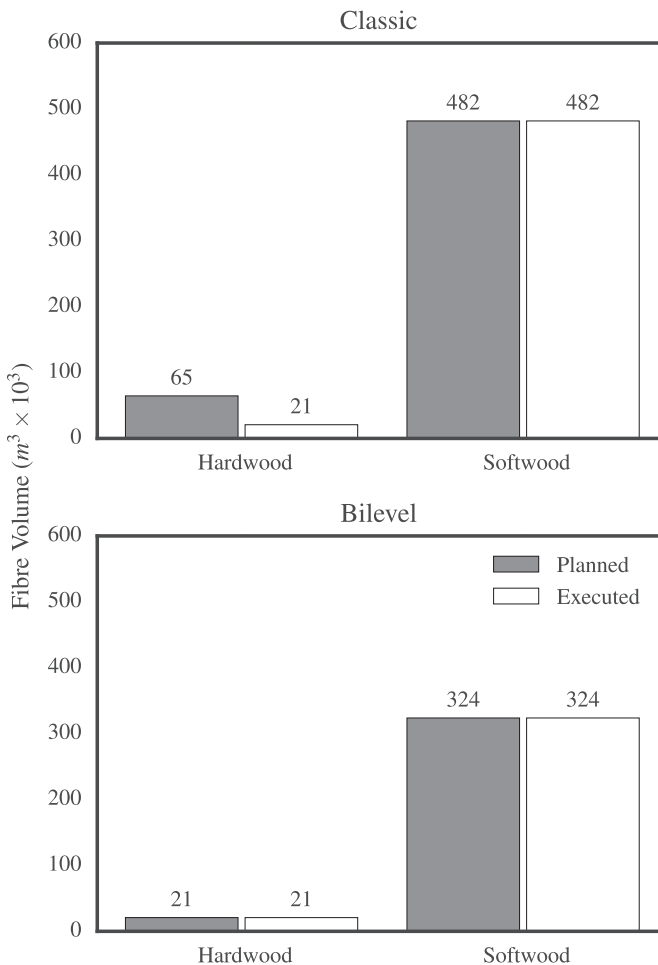
We present experimental results in two stages. Figure 8 and Table 2 compare output from the first planning cycle for scenarios 1 and 3. Figure 9 shows results of simulating 30 sequential rolling-horizon planning cycles for scenarios 1–5.

For scenario 1, the control scenario, potential hardwood fibre supply is 64 583 m³, whereas actual consumption is only 20 800 m³ (Fig. 8). The difference between planned and executed hardwood fibre consumption volumes is due to the limited processing capacity at the (single) hardwood sawmill in the lower-level dataset. The entire softwood fibre supply is consumed by the follower, as we would expect, as both end-product demand and processing capacity for the softwood line are high enough to accommodate more softwood fibre than the forest can supply. This phenomenon can be observed to varying extents in practice, as processing capacity and market demand are often misaligned with the proposed wood supply. To consume the full softwood supply, while only harvesting one-third of the hardwood supply, the follower must harvest stands that have a higher proportion of softwood and lower proportion of hardwood in the second stage of each simulation’s iteration.

For scenario 3 (i.e., the basic bi-level scenario), the anticipation mechanism completely eliminates over-estimation of hardwood fibre consumption. Fibre consumption is exactly equal to wood supply volumes (20 800 m³ for hardwood and 323 759 m³ for softwood; Fig. 8). This is the desired outcome from the bi-level model and represents a global optimal solution for this instance. Note that the follower plans his own harvesting in the second stage of the simulation (using his single-period profit-maximizing model), so harvest areas may not match the upper-level solution, although harvest volume is exactly equal to AAC.

Table 2 presents intermediate results from each step of the bi-level solution method for scenario 3. We present this data as an example of how we derive the optimal upper bounds to the wood supply problem from solutions to output-wise subproblems. Hard-

Fig. 8. Comparison of planned and executed volumes, using classic and bi-level wood supply model formulations.



wood consumption capacity ($20\,800\,m^3$) is the binding constraint in this case. The softwood line could have consumed up to $584\,861\,m^3$ in the first planning period. As expected, fibre volume consumed by the follower in the second phase of the simulation is exactly equal to the wood supply. The bi-level model eliminates the gap between planned and executed fibre consumption levels, thereby fulfilling its intended purpose.

We solve the bi-level model in less than twice the time required to solve the classic model. The classic model can be solved in a single step, which corresponds to approximately 13 seconds of CPU time for our test setup. The bi-level model requires $|O'| + 1$ steps to solve, which corresponds to approximately $(4 + 6) + 10$ seconds of CPU time using our test setup. Note that each rolling-horizon-re-planning simulation scenario requires solution of the model for each period (30 times, for our test case). The solution times reported for classic and bi-level models are averages over the 30 solutions constituting each of scenarios 1 and 3, respectively. We ran our tests on an Intel® Xeon® E5-2670 processor (20 MB cache, 2.60 GHz).

Figure 9 shows sequential re-planning simulation results for the five scenarios described in Section 2.5.3. For each scenario, we plot both species-wise AAC and fibre consumption for each of the 30 rolling-horizon re-planning simulation iterations.

4. Discussion

Scenario 1 shows the relative instability of the classic wood supply model. This is attributable to the species-skewed gap between AAC and fibre consumption volumes. Despite the harvest levels being

Table 2. Bi-level solution method (intermediate results).

Stage	Description	Volume (m^3)	
		Hardwood	Softwood
1 (leader)	Upper bound on hardwood consumption	20 800	—
1 (leader)	Upper bound on softwood consumption	—	584 861
1 (leader)	Maximum even-flow wood supply levels	20 800	323 759
2 (follower)	Fibre consumption	20 800	323 759

systematically lower than AAC, the follower's preference for harvesting stands with high softwood content gradually shifts the composition of the residual forest cover towards higher hardwood content. This undesirable shift in forest composition is not predicted by the leader when using the classic wood supply model. Paradis et al. (2013) use the term systematic drift effect to describe this phenomenon.

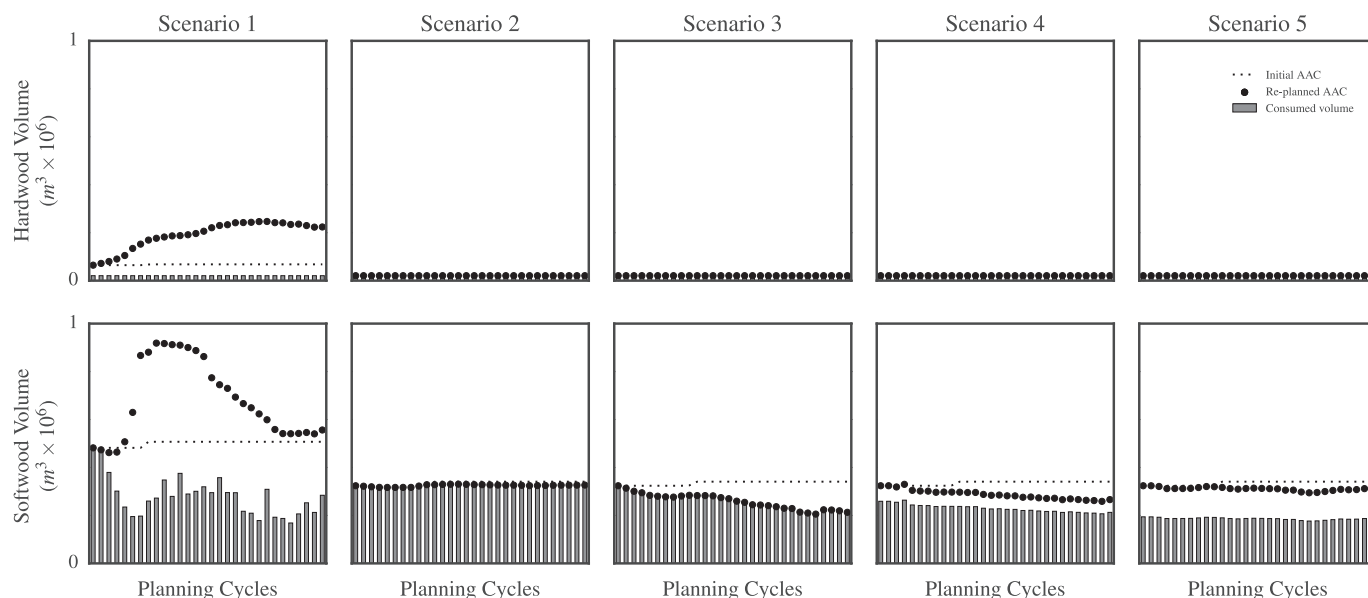
The leader uses the bi-level model to determine AAC for scenarios 2–5. By virtue of its formulation, the bi-level model completely eliminates the volume gap between AAC and fibre consumption in scenarios 2 and 3. Note that we simulate perfect anticipation of industrial fibre consumption volume. For scenarios 3–5, we allow the follower to plan his own harvesting in the second phase of each planning cycle simulation — this explains the residual instability (or drift) in long-term wood supply, relative to initial AAC.

Scenario 2 forces the follower to harvest the exact forest units that form the basis of the first period of the leader's optimal bi-level solution. This scenario simulates the long-term outcome if the optimistic assumption in the OBLPP holds true and represents the best-case outcome of the bi-level planning model. The purpose of this scenario is to show that wood supply tracks almost perfectly along the initial bi-level AAC solution (even after 30 rolling-horizon re-planning cycles) under these (optimistic) best-case conditions (i.e., when the leader controls wood procurement planning and execution all the way to the mill gate). In practice, the decoupling point between leader and follower is not typically located this far downstream.

Scenarios 3–6 simulate the residual drift induced by a more conventional decoupling point, where the follower is allowed some control over first-period harvest planning. Haley and Nelson (2007) discuss alternative tenure systems for managing public forest in Canada, which can impact the decoupling point between government planners and industrial fibre consumers. Better coordination between wood supply planning and industrial fibre consumption planning could potentially reduce pressure related to fibre production on the forest ecosystem, thereby providing some measure of freedom in the solution space for accommodation of nontimber values in the forest management process. Note that in Quebec, the wood supply planning process has recently been decentralised, and regional planning offices have a certain level of autonomy in customizing the formulation of wood supply planning models to accommodate local needs and conditions.

Scenario 3 shows improved softwood supply stability, relative to the control scenario. Note the gradual downward trend of the softwood fibre supply for scenario 3, which contrasts with the more even supply profile simulated in scenario 2. The contrast between scenarios 2 and 3 shows that the bi-level model is sensitive to deviations from the optimal bi-level wood supply model solution. Sensitivity to deviations from the optimal solution is typical of deterministic optimisation models, as optimal solutions are invariably located along the boundary of the feasible region — even the slightest deviations from the optimal solution (or error in constraint values on the right-hand side) can induce problem infeasibility.

Fig. 9. Species-wise bi-level AAC and fibre consumption for scenarios 1–5. The top row shows hardwood volume, whereas the bottom row shows softwood volume. Dotted lines represent initial AAC, circles show re-planned AAC, and bars show volume that was actually consumed by the lower-level model in the second stage of each simulation cycle.



Scenarios 4 and 5 show the effect of reducing the proportion of softwood AAC that is allocated to the follower, in an attempt to compensate for the residual drift seen in scenario 3. Reducing allocation is an indirect way for the leader to induce a buffer stock in the standing timber inventory. This tends to move the executed (second-stage) solution away from the feasible boundary of the planned (first-stage) solution space, thereby improving the robustness of the distributed wood supply planning process. Scenario 4 shows a marked reduction in drift compared with scenario 3. Residual drift is virtually eliminated in scenario 5. Intuitively, withholding 40% of AAC seems like a high penalty to eliminate the residual drift in the bi-level model, and the leader would likely meet with political opposition if he chose to implement such a policy. We conjecture that by using a more direct management approach to maintaining a buffer stock in standing timber inventory, as described in Raulier et al. (2014), it may be possible to achieve higher stable bi-level AAC levels. This represents a promising direction for further research.

By stabilising the long-term wood supply, scenario 5 succeeds in restoring credibility to the wood supply planning process, albeit at a relatively high cost in terms of withheld AAC. Furthermore, scenario 5 makes less optimistic assumptions regarding follower behaviour than scenarios 2, which simulates a perfectly compliant follower. Scenario 5 respects the even-flow pattern prescribed by the classic wood supply model constraints. Assuming that the even-flow constraints are valid and necessary (although not sufficient) conditions for sustainability of the forest management plan and that the leader's responsibility to ensure sustainability must absolutely supersede any desire to maximise short-term TL volume allocations, scenario 5 represents the only example of a sustainable policy in this case study.

Note that if the leader could bundle economically unattractive portions of potential wood supply with more interesting portions, thereby forcing the follower to consume the unattractive portions as a condition on gaining access to the interesting portions, then the antagonism between leader and follower would disappear. This bundling policy option, if it were available to the leader, would change the nature of the problem that we address here, and the bi-level model formulation may no longer be optimal. However, this bundling option would require a more highly constrained contract binding the follower to the leader. If this bundling policy option is

not available to the leader in practice, he has no leverage with which to force the follower to consume economically unattractive fibre. In this case, his only rational choice is to lower the wood supply offer until the follower willingly consumes it all.

Recall that our decomposition methodology converges to a globally optimal solution for special cases (with no saturated capacity constraints on convergent nodes in the lower-level network) and that any problematic general instance can be transformed to an instance of special case 2 by relaxing the saturated capacity constraint just enough to avoid saturation. Alternatively, saturation of the convergent node could be avoided by reducing flow from upstream nodes (i.e., by tightening upstream resource capacity constraints). Finally, our methodology could be used to solve problematic general instances directly (i.e., without any modification of resource capacity constraints), at the expense of guaranteed convergence on a globally optimal solution. For many complex forest management optimization problems (e.g., most spatially explicit multi-period harvest scheduling problems), no solution method guarantees convergence on global optimality (hence the extensive literature on metaheuristic methods for these types of problems). We are assuming that the substantial benefits of anticipating industrial fibre consumption outweigh any potential distortions in the consumption anticipation function induced by relaxing the shared capacity constraint (to convert a problematic general case to special case 2) — we have not, however, conducted an elaborate sensitivity analysis on this particular assumption, as this falls outside the scope of this particular study.

Conceptually, our bi-level formulation would be applicable to a case where the lower-level model includes a stand-level spatially explicit harvest scheduling component (harvest scheduling is currently aspatial, notwithstanding the possibility of augmenting the spatially referenced nature of a given instance by partitioning the forest dataset into a large number of zones, even down to stand level). However, spatially explicit harvest scheduling optimization models (Gunn and Richards 2005; Lockwood and Moore 1993; Martins et al. 2005; Liu et al. 2006; Baskent and Keles 2005) tend to be NP-hard combinatorial problems, which would tend to considerably extend solution times relative to results reported here (possibly beyond the point of practical tractability, although this has not been tested yet). Ideally, the lower-level model in our bi-level framework would provide a perfect anticipation of the fibre-

consumption process, including the exact location and nature of harvest treatments. Extending the method presented here to include a more realistic spatially explicit harvest schedule optimization component would likely further improve the model's ability to mitigate risk of future wood supply shortages, insofar as this refinement of the model more closely reflects the fibre that will actually be harvested in the first period. However, we cannot say, without further testing, the extent to which this additional level of detail would improve the solution (given other sources of uncertainty in the model, which may dominate the solution).

We tested our proposed methodology on a single realistic dataset. The upper-level dataset is compiled from a real (albeit simplified) government wood supply model used to derive AAC. The lower-level dataset is synthetic, to a certain extent, but was compiled from real mill data collected in the course of previous studies undertaken by the FORAC Research Consortium. Although we made every effort to compile a realistic case study, the results presented here are nonetheless derived from a single dataset, which required considerable effort to compile and validate. We recommend further testing on additional datasets prior to deploying the method in a production setting.

Solutions times for both classic and bi-level models were on the order of seconds. Both upper- and lower-level models are linear, so we expect solution times to scale up or down in as an approximately linear function of the complexity of input datasets for upper- and lower-level problems. Given the large spatial and temporal scale of these wood supply models and the fact that they need to be re-solved at periodic (5 or 10 year) intervals, the difference between classic and bi-level model solution times is trivial. Furthermore, we consider accuracy (i.e., absence of bias) in the long-term projections to be much more important than model runtime in this context.

We simulated the DWSPP as a two-stage sequential game, where the leader proposes his wood supply in the first stage and the follower consumes a profit-maximizing subset of the wood supply in the second stage, albeit at a high cost in terms of short-term fibre allocation. At this point, we can conjecture that stable increases in AAC, relative to the bi-level solutions presented here, may be achievable if the leader and the follower were allowed to iteratively adjust their respective supply and demand offers within a given planning cycle. This represents a promising direction for further wood supply policy research. From a game-theoretic perspective, extending the two-stage game simulated in this study to include an iterative negotiation dimension corresponds to a repeated game or supergame in game theory. Under certain conditions, supergames are known to converge on socially optimum equilibrium solutions (i.e., collaborative solutions) that are globally superior to the (optimal) selfish behaviour in the context of nonrepeated (i.e., one-shot) games (Fudenberg and Tirole 1991). The concept of supergames could also be used on a larger scale, to model leader and follower anticipation of upcoming planning cycles (and, potentially, memory of past planning cycles). Ultimately, both scales could be nested (i.e., iterative negotiation within each planning cycle, combined with anticipation of upcoming planning cycles). These hypothetical nested-supergame models might be harnessed for practical application using a metagaming approach (Howard 1971), potentially providing a wealth of valuable insight to guide high-level government policy-makers.

5. Conclusion

Paradis et al. (2013) conjectured that extending the classic wood supply model formulation to anticipate industrial fibre consumption would improve wood supply stability (i.e., mitigate risk of wood supply failure). We framed the DWSPP as a Stackelberg leader-follower game and proposed a bi-level formulation that optimises the outcome from the perspective of the leader. We isolated two special cases of the bi-level problem, which can be

decomposed into output-wise convex subproblems and presented an algorithm that solves the special case of the bi-level problem to global optimality.

We tested our solution methodology on a synthetic dataset of realistic size and complexity, and compared the results with output from the classic (single-level) wood supply optimisation model. Using a series of five scenarios, we showed that the bi-level model improves long-term wood supply stability, although instability is not completely eliminated by the bi-level model when the follower is allowed to re-plan first-period harvesting in the second stage of each planning cycle. We showed that the leader can compensate for this residual instability by withholding (i.e., not attributing) a fraction of bi-level AAC. We conjecture that a similar stabilising effect could be achieved more efficiently (i.e., at a lower cost in terms of withheld bi-level AAC) using a more direct buffer stock modelling approach.

The bi-level model has the same output data format as the classic model and can be solved using comparable computational effort. As such, the bi-level model formulation constitutes a technically compatible and conceptually superior alternative to the classic model.

The current study examines the performance of a bi-level model formulation in the context of a two-stage Stackelberg game. We recommend that research effort on bi-level wood supply model formulations be extended to supergame contexts, in terms of both intra-cycle leader-follower negotiations and inter-cycle anticipation of future wood supply planning games. To cope with the complexity of using these hypothetical nested-supergame models in a practical environment of government policy setting, we suggest the adoption of a metagaming approach to wood supply planning as an appropriate starting point for further study.

Acknowledgements

This study was supported by funding from the FORAC Research Consortium, the Fonds de recherche et de développement en foresterie (FRDF), the Fonds de recherche du Québec — Nature et technologies (FQRNT), the Centre interuniversitaire de recherche sur les réseaux d'entreprise, la logistique, et le transport (CIRRELT), the NSERC Strategic Research Network on Value Chain Optimization (VCO), and NSERC Discovery Grant 203193. The authors thank Frédéric Raulier for his valuable comments and suggestions at various stages in the preparation of this manuscript.

References

- Bard, J.F., and Falk, J.E. 1982. An explicit solution to the multi-level programming problem. *Comput. Oper. Res.* 9(1): 77–100. doi:10.1016/0305-0548(82)90007-7.
- Barros, O., and Weintraub, A. 1982. Planning for a vertically integrated forest industry. *Oper. Res.* 30(6): 1168–1182. doi:10.1287/opre.30.6.1168.
- Baskett, E.Z., and Keles, S. 2005. Spatial forest planning: a review. *Ecol. Modell.* 188(2–4): 145–173. doi:10.1016/j.ecolmodel.2005.01.059.
- Beaudoin, D., Frayret, J.-M., and LeBel, L. 2010. Negotiation-based distributed wood procurement planning within a multi-firm environment. *For. Policy Econ.* 12(2): 79–93. doi:10.1016/j.forpol.2009.07.007.
- Bitran, G.R., and Hax, A.C. 1977. On the design of hierarchical production planning systems. *Decis. Sci.* 8(1): 28–55. doi:10.1111/j.1540-5915.1977.tb01066.x.
- Canadian Council of Forest Ministers. 2003. Defining sustainable forest management in Canada: criteria and indicators. *Can. For. Serv. Tech. Rep. Fo75-3/5-2003E*. Natural Resources Canada, Canadian Forest Service.
- Canadian Council of Forest Ministers. 2008. A vision for Canada's forests: 2008 and beyond. *Can. For. Serv. Tech. Rep. Fo4-23/2008E-PDF*. Natural Resources Canada, Canadian Forest Service.
- Canadian Council of Forest Ministers. 2014. National Forestry Database [online]. Available from <http://nfdp.ccfm.org> [accessed 1 September 2014].
- Colson, B., Marcotte, P., and Savard, G. 2007. An overview of bilevel optimization. *Ann. Oper. Res.* 153(1): 235–256. doi:10.1007/s10479-007-0176-2.
- Davis, L.S., Johnson, K.N., Bettinger, P., and Howard, T.E. 2001. *Forest management: to sustain ecological, economic, and social values*. McGraw Hill, New York.
- Dempe, S., Kalashnikov, V., Pérez-Valdés, G.A., and Kalashnykova, N. 2015. *Bi-level programming problems: theory, algorithms and applications to energy networks*. Springer-Verlag, Berlin Heidelberg. doi:10.1007/978-3-662-45827-3.
- Fortuny-Amat, J., and McCarl, B. 1981. A representation and economic interpretation

- tation of a two-level programming problem. *J. Oper. Res. Soc.* **32**(9): 783–792. doi:[10.1057/jors.1981.156](https://doi.org/10.1057/jors.1981.156).
- Fudenberg, D., and Tirole, J. 1991. *Game theory*. MIT Press, Cambridge, Mass.
- Gunn, E.A. 2007. Models for strategic forest management. In *Handbook of operations research in natural resources*. Edited by A. Weintraub, C. Romero, T. Bjørndal, and R. Epstein. Springer, Boston. Chapter 16, pp. 317–341. doi:[10.1007/978-0-387-71815-6_16](https://doi.org/10.1007/978-0-387-71815-6_16).
- Gunn, E.A., and Rai, A.K. 1987. Modelling and decomposition for planning long-term forest harvesting in an integrated industry structure. *Can. J. For. Res.* **17**(12): 1507–1518. doi:[10.1139/x87-233](https://doi.org/10.1139/x87-233).
- Gunn, E.A., and Richards, E.W. 2005. Solving the adjacency problem with stand-centred constraints. *Can. J. For. Res.* **35**(4): 832–842. doi:[10.1139/x05-013](https://doi.org/10.1139/x05-013).
- Haley, D., and Nelson, H. 2007. Has the time come to rethink Canada's Crown forest tenure systems? *For. Chron.* **83**(5): 630–641. doi:[10.5558/tfc83630-5](https://doi.org/10.5558/tfc83630-5).
- Howard, N. 1971. *Paradoxes of rationality: theory of metagames and political behavior*. MIT Press, Cambridge, Mass.
- Jerbi, W., Gaudreault, J., D'Amours, S., Nourelfath, M., Lemieux, S., Marier, P., and Bouchard, M. 2012. Optimization/simulation-based framework for the evaluation of supply chain management policies in the forest product industry. In *2012 IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, Seoul, South Korea, 14–17 October 2012. Institute of Electrical and Electronics Engineers, Piscataway, N.J. doi:[10.1109/ICSMC.2012.6377989](https://doi.org/10.1109/ICSMC.2012.6377989).
- Lammerts van Bueren, E.M., and Blom, E.M. 1997. Hierarchical framework for the formulation of sustainable forest management standards. Tropenbos International, Wageningen, the Netherlands.
- Liu, G., Han, S., Zhao, X., Nelson, J.D., Wang, H., and Wang, W. 2006. Optimisation algorithms for spatially constrained forest planning. *Ecol. Modell.* **194**(4): 421–428. doi:[10.1016/j.ecolmodel.2005.10.028](https://doi.org/10.1016/j.ecolmodel.2005.10.028).
- Lockwood, C., and Moore, T. 1993. Harvest scheduling with spatial constraints: a simulated annealing approach. *Can. J. For. Res.* **23**(3): 468–478. doi:[10.1139/x93-065](https://doi.org/10.1139/x93-065).
- Martins, I., Constantino, M., and Borges, J.G. 2005. A column generation approach for solving a non-temporal forest harvest model with spatial structure constraints. *Eur. J. Oper. Res.* **161**(2): 478–498. doi:[10.1016/j.ejor.2003.07.021](https://doi.org/10.1016/j.ejor.2003.07.021).
- Mathey, A.-H., Nelson, H., and Gaston, C. 2009. The economics of timber supply: does it pay to reduce harvest levels? *For. Policy Econ.* **11**(7): 491–497. doi:[10.1016/j.forpol.2009.05.006](https://doi.org/10.1016/j.forpol.2009.05.006).
- Paradis, G. 2016. Hierarchical forest management planning: a bilevel wood supply modelling approach. Ph.D. thesis, Université Laval, Quebec City, Que.
- Paradis, G., LeBel, L., D'Amours, S., and Bouchard, M. 2013. On the risk of systematic drift under incoherent hierarchical forest management planning. *Can. J. For. Res.* **43**(5): 480–492. doi:[10.1139/cjfr-2012-0334](https://doi.org/10.1139/cjfr-2012-0334).
- Raulier, F., Dhital, N., Racine, P., Tittler, R., and Fall, A. 2014. Increasing resilience of timber supply: how a variable buffer stock of timber can efficiently reduce exposure to shortfalls caused by wildfires. *For. Policy Econ.* **46**: 47–55. doi:[10.1016/j.forpol.2014.06.007](https://doi.org/10.1016/j.forpol.2014.06.007).
- Schneeweiss, C. 2003. *Distributed decision making*. Springer-Verlag, Berlin Heidelberg. doi:[10.1007/978-3-540-24724-1](https://doi.org/10.1007/978-3-540-24724-1).
- Tittler, R., Messier, C., and Burton, P.J. 2001. Hierarchical forest management planning and sustainable forest management in the boreal forest. *For. Chron.* **77**(6): 998–1005. doi:[10.5558/tfc77998-6](https://doi.org/10.5558/tfc77998-6).
- von Stackelberg, H. 2011. *Market structure and equilibrium*. Springer-Verlag, Berlin Heidelberg. doi:[10.1007/978-3-642-12586-7](https://doi.org/10.1007/978-3-642-12586-7).
- Weintraub, A., and Cholak, A. 1991. A hierarchical approach to forest planning. *For. Sci.* **37**(2): 439–460.
- Weintraub, A., and Davis, L. 1996. Hierarchical planning in forest resource management: defining the dimensions of the subject area. In *Proceedings of a Workshop: Hierarchical Approaches to Forest Management in Public and Private Organizations*, Toronto, Ont., 25–29 May 1992. Edited by D.L. Martell, L.S. Davis, and A. Weintraub. Can. For. Serv. Petawawa Natl. For. Inst. Inf. Rep. PI-X-124. Canadian Forestry Service. pp. 2–14.
- Weintraub, A., Romero, C., Bjørndal, T., and Epstein, R. (Editors). 2007. *Handbook of operations research in natural resources*. Springer, Boston. doi:[10.1007/978-0-387-71815-6](https://doi.org/10.1007/978-0-387-71815-6).